2025_MCET_Soliton_Mock Test 2

Test Summary

- No. of Sections: 3
- No. of Questions: 19
- Total Duration: 105 min

Section 1 - Physics and Core MCQs

Section Summary

- No. of Questions: 10
- Duration: 10 min

Additional Instructions:

None

Q1. When the length of the conductor is doubled and the area of cross-section remains the same then its resistance

- Remains the same
- Will be doubled
- Will become half
- Will increase by four times

Q2. Which of the following logic gate is a universal gate?

- OR
- NOT
- AND
- NOR

Q3. Identify which one is a first order phase transition?

- A liquid to gas transition at its crucial temperature.
- A liquid to gas transition close to its triple point.
- A paramagnetic to ferromagnetic transition in the absence of s magnetic field.
- None of the above.

Q4. Which phenomenon is responsible for the electrical conductivity in metals?

- Photoelectric effect
- Superconductivity
- Free electron model
- Band theory

Q5. How does the period of oscillation change if the spring constant of a mass-spring system is doubled?

- The period is halved
- The period is doubled
- The period remains the same
- The period becomes four times

Q6. What is the principle of virtual work in the context of static equilibrium?

- The work done by external forces is zero
- The work done by internal forces is zero
- The virtual displacement is always zero
- The work done by friction is zero

Q7. What is the function of a coronagraph in solar physics?

- To measure solar wind velocity
- To observe solar flares
- To block out the Sun's bright disk and study its outer atmosphere
- To capture images of sunspots

Q8. Find the value of a that satisfies the equation $2^{3a-1}=8$.

- a=2

a=1

a=3

a=4

Q9. The slope of the line passing through the points (2, 3) and (5, 7) is:

0

1

2

3

Q10. Find the slope of the line passing through the points (2,5) and (4,9).

2

3

4

1/2

Section 2 - Technical MCQs

Section Summary

- No. of Questions: 5
- Duration: 5 min

Additional Instructions:

None

Q1. What will be the output of the following program?

```
#include <stdio.h>
int main() {
  int a;
  int b = 1;
  int x[5] = { 1, 2, 3, 4, 5 };
  a = 5 * 4 + x[b++] - (9 / b);
  printf("%d", a);
  return 0;
}
```

12

20

18

10

Q2. Comment on the following program.

```
#include <stdio.h>
int main() {
   int i;
   int arr[5] = {0};
   for (i = 0; i <= 5; i++)
      printf("%d", arr[i]);
   return 0;
}
```

The program always crashes

The program always prints 0 six times followed by garbage value

Compiler Error: Array index out of bound

The program may print 0 five times followed by garbage value, or may crash if address (arr+5) is invalid

Q3. What will be the output of the following program?

```
#include<stdio.h>
int main() {
   float arr[]={12.4, 2.3, 4.5, 6.7};
   printf("%d", sizeof(arr)/sizeof(arr[0]));
   return 0;
}
```

5

4

16

7

Q4. What will be the output of the following program?

```
#include <stdio.h>
int main() {
   char a[2][6] = {"hello", "hi"};
   printf("%d", sizeof(a));
   return 0;
}
```

9

10

11

12

Q5. What will be the output of the following program?

```
#include<stdio.h>
#include<string.h>

int main() {
   char str[][15] ={ "2DCharArray", "2DStrings"};
   printf("%c", str[1][5]);

   return 0;
}
```

r

i

n

S

Section 3 - Coding

Section Summary

- No. of Questions: 4
- Duration: 90 min

Additional Instructions:

None

Q1. **Problem Statement**

Bob, an enthusiastic coding seminar attendee, has a task that requires finding the frequency of a specific character in a given string.

Your task is to assist Bob in accomplishing this. Help him find the frequency of a character in a given string by utilizing character arithmetic.

Input Format

The first line of input consists of a string of printable ASCII characters, including spaces.
The second line of input consists of a single character for which the frequency needs to be determined.

Output Format

The output prints an integer, representing the frequency (count) of the specified character in the given string.
If there is no specified character, it prints "0".

Refer to the sample outputs for the formatting specifications.

Constraints

In this scenario, the test cases fall under the following constraints:
5 ≤ length of string ≤ 49
The character to be counted is not case-sensitive.

Sample Input

```
Sample Data for analysis
a
```

Sample Output

```
5
```

Sample Input

```
Hello Learn Programming
z
```

Sample Output

```
0
```

Sample Input

```
Hello
L
```

Sample Output

```
2
```

Sample Input

```
There are three Different colors Red, Blue and Grd
d
```

Sample Output

```
4
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

Harpreet is a student learning about matrix multiplication. She wants to write a program to multiply two matrices, **a** and **b**, each of size N x N. Can you help her with the code?

Write a program that takes an integer N as input and two matrices **a** and **b** of size **N x N**. Implement matrix multiplication and print the resulting matrix **c**.

Note: A square matrix is where the number of rows equals the number of columns.

Input Format

The first line of input consists of an integer **N,** representing the matrix size.
The next **N** lines consist of **N** space-separated elements in each line, representing the first matrix.
After being separated by a new line, the next **N** lines consist of **N** space-separated elements in each line, representing the second matrix.

Output Format

The output prints the product of two matrices.

Refer to the sample output for the formatting specifications.

Constraints

In the given scenario, the test cases fall under the following constraints:
2 ≤ N ≤ 8
0 ≤ elements ≤ 9

Sample Input

```
3
2 3 2
3 2 3
3 3 3
```

Sample Output

```
16 23 21
19 27 29
21 30 30
```

Sample Input

```
2
2 2
```

Sample Output

```
24 16
31 24
```

```
2 3
```

Sample Input

```
8
1 5 2 4 7 3 2 1
2 3 4 5 6 7 8 9
4 5 6 1 2 3 7 8
```

Sample Output

```
99 131 152 141 106 88 97 115
145 200 236 248 183 155 137 181
155 177 179 207 172 108 88 113
157 201 217 233 202 166 146 169
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q3. **Problem Statement**

Write a program to get **N** integer inputs in an array.

Select an element from the array and count the following:

1. The number of elements that are greater than the selected element
2. The number of elements that are lesser than the selected element
3. The number of elements that are exactly divisible by the selected element

Input Format

The first line of input consists of an integer **N,** representing the size of the array.
The second line consists of **N** elements of the array, separated by space.
The third line consists of the selected element from the array.

Output Format

The first line of output prints the number of elements greater than the selected element.
The second line prints the number of elements less than the selected element.
The third line prints the number of elements that are exactly divisible by the selected element.

Refer to the sample output for formatting specifications.

Constraints

$2 \leq N \leq 30$

Sample Input

```
12
12 23 45 56 78 89 98 87 65 54 32 21
32
```

Sample Output

```
Greater : 8
Lesser : 3
Exactly divisible : 1
```

Sample Input

```
5
1 3 4 5 2
2
```

Sample Output

```
Greater : 3
Lesser : 1
Exactly divisible : 2
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q4. Write a program to find the length of the given input string and print the reverse of the given string without using library functions.

Input Format

The input consists of a string str.

Output Format

The first line prints the length of the string.
The second line prints the reversed string.

Refer to the sample input and output for formatting specifications.

Constraints

1 <= str.length() <= 100

Sample Input

```
hello
```

Sample Output

```
The length of the string is: 5
The reverse of the string is: olleh
```

Sample Input

```
sample
```

Sample Output

```
The length of the string is: 6
The reverse of the string is: elpmas
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Physics and Core MCQs

Q1

Will be doubled

Solution

No Solution

Q2

NOR

Solution

No Solution

Q3

A liquid to gas transition at its crucial temperature.

Solution

No Solution

Q4

Free electron model

Solution

No Solution

Q5

The period remains the same

Solution

No Solution

Q6

The work done by internal forces is zero

Solution

No Solution

Q7

To block out the Sun's bright disk and study its outer atmosphere

Solution

No Solution

Q8

a=1

Solution

No Solution

Q9

2

Solution

No Solution

Q10

3

Solution

No Solution

Section 2 - Technical MCQs

Q1

18

Solution

No Solution

Q2

The program may print 0 five times followed by garbage value, or may crash if address (arr+5) is invalid

Solution

No Solution

Q3

4

Solution

No Solution

Q4

12

Solution

No Solution

Q5

i

Solution

No Solution

Section 3 - Coding

Q1

Hidden Test Case

Input	Output
This is a good platform to learn 1	2

Weightage - 15

Input

```
coding makes learning easy
k
```

Output

```
1
```

Weightage - 15

Input

```
Join the programming session
z
```

Output

```
0
```

Weightage - 10

Input

```
Follow your Heart and Brain
r
```

Output

```
3
```

Weightage - 10

Input

```
PPPP pPpP asdfghjkl PPpp
P
```

Output

```
12
```

Weightage - 25

Input

```
This is a Hard-Level Test Case with several chara
t
```

Output

```
5
```

Weightage - 25

Sample Input

```
Sample Data for analysis
a
```

Sample Output

```
5
```

Sample Input

```
Hello Learn Programming
z
```

Sample Output

```
0
```

Sample Input

```
Hello
L
```

Sample Output

```
2
```

Sample Input

```
There are three Different colors Red, Blue and Gr
d
```

Sample Output

```
4
```

Solution

```c
#include <stdio.h>

int main() {
    char str[50], ch;
    int count = 0;

    scanf("%[^\n]s", str);
    scanf(" %c", &ch);

    for (int i = 0; str[i] != '\0'; ++i) {
        if ((ch >= 'A' && ch <= 'Z' && (ch == str[i] || ch + ('a' - 'A') == str[i])) ||
            (ch >= 'a' && ch <= 'z' && (ch == str[i] || ch - ('a' - 'A') == str[i]))) {
            ++count;
        }
    }

    printf("%d", count);
    return 0;
}
```

Q2

Hidden Test Case

Input

```
4
1 2 3 4
7 8 9 6
4 5 6 3
```

Output

```
46 56 42 27
162 162 126 91
96 93 72 55
24 33 24 15
```

Weightage - 15

Input

```
5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

Output

```
60 75 90 0 15
60 75 90 0 15
60 75 90 0 15
60 75 90 0 15
```

Weightage - 15

Input

```
3
7 8 1
9 4 3
15 7 5
```

Output

```
16 47 7
8 29 9
14 50 15
```

Weightage - 10

Input

```
8
1 2 3 4 5 6 7 8
2 3 4 5 6 7 8 9
```

Output

```
174 210 183 174 183 210 174 156
210 253 224 213 220 254 216 196
183 215 247 216 203 226 249 209
```

Weightage - 25

Input

```
2
9 6
6 9
```

Output

```
63 78
57 87
```

Weightage - 10

Input

```
7
7 8 9 4 5 6 1
2 5 8 7 4 1 3
```

Output

```
87 176 174 236 167 241 197
65 107 118 156 138 187 156
59 102 110 168 145 196 164
```

Weightage - 25

Sample Input

```
3
2 3 2
3 2 3
```

Sample Output

```
16 23 21
19 27 29
21 30 30
```

Sample Input

```
2
2 2
2 3
```

Sample Output

```
24 16
31 24
```

Sample Input

```
8
1 5 2 4 7 3 2 1
2 3 4 5 6 7 8 9
```

Sample Output

```
99 131 152 141 106 88 97 115
145 200 236 248 183 155 137 181
155 177 179 207 172 108 88 113
```

Solution

```
#include<stdio.h>
int main() {
    int a[8][8], b[8][8], c[8][8], n, i, j, k;
    scanf("%d", & n);

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", & a[i][j]);
        }
    }


    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", & b[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            c[i][j] = 0;
            for (k = 0; k < n; k++) {
```

```
            c[i][j] += a[i][k] * b[k][j];
        }
    }
}

for (i = 0; i < n; i++) {
    for (j = 0; j < n; j++) {
        printf("%d ", c[i][j]);
    }
    printf("\n");
}
return 0;
}
```

Q3

Hidden Test Case

Input

```
10
12 13 30 32 45 21 24 14 23 18
45
```

Output

```
Greater : 0
Lesser : 9
Exactly divisible : 1
```

Weightage - 10

Input

```
8
21 23 24 32 34 36 38 40
32
```

Output

```
Greater : 4
Lesser : 3
Exactly divisible : 1
```

Weightage - 10

Input

```
12
12 23 45 56 78 89 98 87 65 54 32 21
2
```

Output

```
Greater : 12
Lesser : 0
Exactly divisible : 6
```

Weightage - 15

Input

```
20
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
25
```

Output

```
Greater : 0
Lesser : 20
Exactly divisible : 0
```

Weightage - 25

Input

```
18
1 4 7 8 5 2 3 6 9 14 25 74 85 96 35 19 10 76
2
```

Output

```
Greater : 16
Lesser : 1
Exactly divisible : 9
```

Weightage - 15

Input

Output

```
30
10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 2
4
```

```
Greater : 30
Lesser : 0
Exactly divisible : 7
```

Weightage - 25

Sample Input

```
12
12 23 45 56 78 89 98 87 65 54 32 21
32
```

Sample Output

```
Greater : 8
Lesser : 3
Exactly divisible : 1
```

Sample Input

```
5
1 3 4 5 2
2
```

Sample Output

```
Greater : 3
Lesser : 1
Exactly divisible : 2
```

Solution

```c
#include<stdio.h>
int main()
{
    int i,less=0,greater=0,input,count=0,n;
    scanf("%d",&n);
    int arr[n];
    for(i=0;i<n;i++)
    scanf("%d",&arr[i]);
    scanf("%d",&input);
    for(i=0;i<n;i++)
    {
        if(input<arr[i])
        greater++;
    }
    for(i=0;i<n;i++)
    {
        if(input>arr[i])
        less++;
    }
    for(i=0;i<n;i++)
    {
        if(arr[i]%input == 0)
        count++;
    }
    printf("Greater : %d\n", greater);
    printf("Lesser : %d\n", less);
    printf("Exactly divisible : %d",count);
    return 0;
}
```

Q4

Hidden Test Case

Input

```
flippant
```

Output

```
The length of the string is: 8
The reverse of the string is: tnappilf
```

Weightage - 10

Input

```
exuberant
```

Output

```
The length of the string is: 9
The reverse of the string is: tnarebuxe
```

Weightage - 10

Input

```
HarryPotterAndChamberOfSecrets
```

Output

```
The length of the string is: 30
The reverse of the string is: sterceSfOrebmahCdnAr
```

Weightage - 15

Input

```
Sample+Input+testcase+Answer+solution
```

Output

```
The length of the string is: 37
The reverse of the string is: noitulos+rewsnA+esac
```

Weightage - 15

Input

```
wsedrtfgyhujikolESDRTFGYHUJIKO1234567!@#$%^&*SDFGHJK
```

Output

```
The length of the string is: 52
The reverse of the string is: KJHGFDS*&^%$#@!76543
```

Weightage - 25

Input

```
zxcvbnmasdfghjkqwertyuioWERTYUISDFGHJKZXCVBNM1234567
```

Output

```
The length of the string is: 72
The reverse of the string is: ybvtcrx)(*&^%$#@!098
```

Weightage - 25

Sample Input

```
hello
```

Sample Output

```
The length of the string is: 5
The reverse of the string is: olleh
```

Sample Input

```
sample
```

Sample Output

```
The length of the string is: 6
The reverse of the string is: elpmas
```

Solution

```c
#include <stdio.h>

int main() {
    char str[100];
    int length = 0;
```

```
   // printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    // Find the length of the string
    for (int i = 0; str[i] != '\0'; i++) {
        if (str[i] == '\n') {
            str[i] = '\0'; // Remove the newline character
            break;
        }
        length++;
    }
    printf("The length of the string is: %d\n", length);
    printf("The reverse of the string is: ");
    // Print the reverse of the string
    for (int i = length - 1; i >= 0; i--) {
        printf("%c", str[i]);
    }

      return 0;

}
```